

Daffodil International University (DIU)

Department of Computer Science & Engineering (CSE)

Quiz Examination – Summer 2026

Course Code: CSE 311

Course Title: Database Management Systems (DBMS)

Time: 40 Minutes

Total Marks: 15

Part A: Normalization (7 Marks)

Scenario:

A hospital stores patient appointment and treatment information in a single table as shown below.

PatientID	Patient Name	Age	DoctorID	Doctor Name	Department	TreatmentID	TreatmentName	AppointmentDate	RoomNo
P101	Ayesha	25	D01	Dr. Rahman	Cardiology	T01	ECG	10-07-2026	201
P101	Ayesha	25	D02	Dr. Hasan	Neurology	T02	MRI Scan	15-07-2026	305
P102	Karim	40	D01	Dr. Rahman	Cardiology	T01	ECG	11-07-2026	201
P103	Nabila	30	D03	Dr. Islam	Orthopedics	T03	X-Ray	12-07-2026	102

Questions:

1. Convert the given table into First Normal Form (1NF). (2 Marks)
2. Convert the table into Second Normal Form (2NF) by removing partial dependencies. (2 Marks)
3. Convert the table into Third Normal Form (3NF) by removing transitive dependencies. (3 Marks)

Part B: SQL Queries (8 Marks)

Scenario:

A company maintains the following database:

Employee(EmpID, EmpName, DepartmentID, Salary)

Department(DepartmentID, DepartmentName, ManagerID)

Manager(ManagerID, ManagerName)

- Q1. (2 Marks) Display the Employee Name and Department Name of all employees.
- Q2. (2 Marks) Display the Employee Name, Department Name, and Manager Name for every employee.
- Q3. (2 Marks) Display the Department Name and the total number of employees working in each department. Show only those departments having more than one employee.
- Q4. (2 Marks) Display the Manager Name, Department Name, and the average salary of employees working under each manager. Show only those managers whose team's average salary is greater than 60,000.

Instructions

- Write all SQL keywords in UPPERCASE.
- Assume all PK/FK relationships are correctly defined.
- Write clear and syntactically correct SQL queries.

Best of Luck!